

IRON STUDIES BENCH

TSAT% = $\text{Fe} \div \text{TIBC} \times 100$ · NORMAL 25-50%

APR!

IDA

LOW

DIURNAL

LOW

IN INFLAMMATION USE <100

GREEN >200

YELLOW <100

RED <30

CHF/CKD: USE <20%

SUSPECT LINEUP

HIGH

DOWN THE MIDDLE

sTfR/Ft TOOL >2

SHORT

FULL

NO Fe

PERFECTLY UNIFORM HEIGHT WITH NO NO VARIATION AT ALL

SUSPECT 1

SUSPECT 2

DOWN THE MIDDLE

SHORT

ADVANCED MARKERS SHELF

EARLIEST

CHr

Measures the the newest reticulocytes

%HYPO >6%

Zn-PP

ZINC

INSERTED INTO HEME INSTEAD OF IRON

>80

sTfR/log Ft

>2

>2

<1

BEST FOR MIXED

BM STAIN

GOLD STD - RARE

IRON STUDIES BENCH

SUSPECT LINEUP

ADVANCED MARKERS SHELF

1. TSAT formula

WHAT YOU SEE

Blackboard: $\text{TSAT}\% = \text{Fe} \div \text{TIBC} \times 100$, normal 25–50%

WHAT IT ENCODES

Transferrin saturation = $(\text{serum iron} \div \text{TIBC}) \times 100$; normal 25–50%. It estimates how 'full' the transferrin trucks are. A LOW TSAT (<20%) signals iron-deficient erythropoiesis (inadequate iron delivery to the marrow), whether from true deficiency or functional sequestration.

BOARD TRAP

WRONG: using TSAT alone to diagnose IDA. TSAT is low in BOTH iron deficiency AND anemia of chronic disease, so it cannot separate them by itself. Pair it with ferritin and TIBC: IDA = low ferritin + HIGH TIBC; ACD = high/normal ferritin + LOW TIBC.

2. Iron scale / TIBC

WHAT YOU SEE

Green-zone scale weighing iron against TIBC capacity

WHAT IT ENCODES

In IDA the liver upregulates transferrin to scavenge scarce iron, so TIBC is HIGH; in ACD, inflammation (hepcidin/IL-6) lowers transferrin and traps iron, so TIBC is LOW and ferritin is high/normal. TIBC moving up = the body is hungry for iron.

BOARD TRAP

WRONG: thinking TIBC rises in every anemia. TIBC is HIGH only in true iron deficiency; it is LOW in anemia of chronic disease, malnutrition, and liver disease. A LOW TIBC in an anemic patient argues AGAINST simple iron deficiency and toward ACD.

3. <20% TSAT dial

WHAT YOU SEE

Gauge red zone <20%; note 'CHF/CKD use <20%'

WHAT IT ENCODES

TSAT <20% defines iron-deficient erythropoiesis. In CKD and CHF, FUNCTIONAL iron deficiency is treated using TSAT <20% (often combined with ferritin <100, or ferritin 100–300 with TSAT <20% in HF) to start IV iron — because hepcidin traps stores even when ferritin looks adequate.

BOARD TRAP

WRONG: waiting for a low ferritin before treating iron deficiency in CKD/HF. In these inflammatory states ferritin is falsely elevated; the operative trigger is TSAT <20% (with the ferritin caveats above), and the treatment is IV iron, not oral.

4. Suspect 1 — IDA

WHAT YOU SEE

Pale bride with green check, ruler showing she is empty/low

WHAT IT ENCODES

Iron deficiency anemia = TRULY empty iron stores. Classic panel: LOW ferritin (<30, most specific), low serum iron, HIGH TIBC, LOW TSAT (<20%), HIGH RDW, microcytic/hypochromic (low MCV/MCH), often reactive thrombocytosis. Mentzer index (MCV/RBC) >13 favors IDA.

BOARD TRAP

WRONG: dismissing a low ferritin as 'just inflammation.' Ferritin is the most specific marker of iron deficiency; a LOW ferritin is diagnostic of IDA even in a sick patient. Inflammation can only falsely RAISE ferritin — it cannot make it falsely low, so a low value is real.

5. Suspect 2 — ACD

WHAT YOU SEE

Hunched figure with red X by a fire (inflammation), 'FULL' stores

WHAT IT ENCODES

Anemia of chronic disease (anemia of inflammation): iron is present but SEQUESTERED by high hepcidin. Labels: ferritin HIGH/normal, TIBC LOW, serum iron LOW, TSAT low, hepcidin high. Usually normocytic but can be mildly microcytic. Underlying chronic infection, autoimmune disease, or malignancy.

BOARD TRAP

WRONG: giving oral iron to an ACD patient and expecting a response. High hepcidin blocks duodenal absorption, so oral iron fails; the stores are full anyway. Treat the underlying disease; use IV iron or ESAs only when a true coexisting deficiency is confirmed (e.g., low sTfR index).

6. Suspect 3 — thalassemia

WHAT YOU SEE

Figure labeled 'sTfR/Ft >2', 'Down the middle', 'FULL', No Fe deficit

WHAT IT ENCODES

Thalassemia trait: a globin-chain production defect, NOT an iron problem. Iron studies are NORMAL (ferritin/iron normal or high). Hallmark: marked microcytosis OUT OF PROPORTION to mild anemia, near-normal RBC COUNT, Mentzer index (MCV/RBC) <13, target cells. Beta-thal confirmed by HbA2 >3.5% on electrophoresis.

BOARD TRAP

WRONG: giving iron to a patient with thalassemia trait because they are microcytic. Iron stores are normal/full — iron does nothing and risks overload over time. The discriminators: NORMAL ferritin, NORMAL/low RDW, normal-to-high RBC count, and family history; confirm with hemoglobin electrophoresis.

7. sTfR/Ferritin >2

WHAT YOU SEE

Sign 'sTfR/Ft >2' over center suspects

WHAT IT ENCODES

Soluble transferrin receptor (sTfR) is shed in proportion to cellular iron HUNGER and, unlike ferritin, is NOT an acute-phase reactant. The sTfR/log-ferritin index is HIGH in IDA and NORMAL in pure ACD, so a ratio >2 (or >1–2 depending on assay) unmasks iron deficiency hidden behind inflammation.

BOARD TRAP

WRONG: expecting sTfR to be elevated in ACD. In pure anemia of chronic disease sTfR is NORMAL (stores are adequate). A HIGH sTfR/log-ferritin index points to true iron deficiency — including IDA coexisting with inflammation. Thalassemia can also raise sTfR (high RBC turnover), so interpret with the full picture.

8. Suspect 4 — sideroblastic

WHAT YOU SEE

Knight in full armor 'No Fe deficit', short, uniform height, 'FULL'

WHAT IT ENCODES

Sideroblastic anemia: iron is ABUNDANT but cannot be incorporated into heme, so it accumulates in mitochondria → RINGED SIDEROBLASTS on Prussian-blue marrow stain. Iron, ferritin, and TSAT are HIGH. Causes: hereditary (X-linked ALAS2), lead poisoning, alcohol, isoniazid/B6 deficiency, copper deficiency/zinc excess, MDS (RARS).

BOARD TRAP

WRONG: confusing sideroblastic anemia with IDA because both can be microcytic. In sideroblastic anemia iron is HIGH, not low — giving iron worsens overload. The lead/INH/alcohol history, high ferritin/TSAT, basophilic stippling (lead), and ringed sideroblasts on marrow are the discriminators; B6 (pyridoxine) treats the reversible forms.

9. Short / uniform RDW clue

WHAT YOU SEE

Caption: perfectly uniform height, no variation at all

WHAT IT ENCODES

RDW measures red-cell size variation (anisocytosis). Thalassemia produces UNIFORMLY small cells → normal/low RDW. IDA produces a mix of normal and small cells → HIGH RDW. So in a microcytic patient, high RDW favors IDA; normal RDW favors thalassemia trait.

BOARD TRAP

WRONG: expecting a HIGH RDW in thalassemia trait. RDW is characteristically NORMAL/low in thalassemia (uniform microcytes) — a high RDW points away from thal and toward iron deficiency. RDW is a cheap, fast first discriminator before ordering ferritin/electrophoresis.

10. sTfR / log-ferritin index

WHAT YOU SEE

Advanced markers shelf: sTfR / log-Ft ratio, 'best when mixed'

WHAT IT ENCODES

The sTfR/log-ferritin index is the BEST test when IDA and ACD COEXIST (mixed picture), where ferritin is uninterpretable. A high index (commonly >2) favors a true iron-deficiency component requiring repletion; a low index suggests pure inflammatory anemia.

BOARD TRAP

WRONG: relying on ferritin alone in an inflamed patient to exclude IDA. Inflammation inflates ferritin and can mask deficiency; the sTfR-based index (sTfR is not acute-phase) is what reliably detects coexisting IDA. Bone-marrow iron stain is the alternative gold standard if still unclear.

11. Reticulocyte Hb (CHr)

WHAT YOU SEE

Shelf marker 'CHr' measuring iron in new reticulocytes

WHAT IT ENCODES

Reticulocyte hemoglobin content (CHr / Ret-He) reflects iron available to the marrow OVER THE LAST FEW DAYS — an early, real-time marker of functional iron deficiency that drops before ferritin/Hb changes and rebounds within days of effective iron therapy. A low CHr (<28–29 pg) suggests iron-restricted erythropoiesis.

BOARD TRAP

WRONG: thinking ferritin/MCV are the earliest signals of iron-restricted erythropoiesis. CHr/Ret-He changes BEFORE the CBC indices and is especially useful in CKD/inflammation where ferritin is unreliable — a normal MCV does not exclude functional iron deficiency.

12. Zinc protoporphyrin

WHAT YOU SEE

Shelf: Zn-PP, iron inserted with no Fe instead → ZPP

WHAT IT ENCODES

In the last step of heme synthesis, ferrochelatase normally inserts iron into protoporphyrin IX. When iron is unavailable (IDA) OR ferrochelatase is inhibited (LEAD poisoning), ZINC is inserted instead → elevated zinc protoporphyrin (ZPP). High ZPP therefore appears in both iron deficiency and lead toxicity.

BOARD TRAP

WRONG: expecting a NORMAL ZPP in lead poisoning. Lead inhibits ferrochelatase (and ALA dehydratase), so ZPP is HIGH in lead toxicity — just as in IDA. Distinguish lead by exposure history, basophilic stippling, and a blood lead level; do not let an elevated ZPP fool you into diagnosing simple IDA.

13. Bone marrow iron stain

WHAT YOU SEE

Shelf bottom: 'BM STAIN — GOLD STD MARROW'

WHAT IT ENCODES

Prussian-blue (Perls) staining of a bone-marrow aspirate is the GOLD STANDARD for assessing iron stores: ABSENT stainable iron confirms IDA, abundant iron with RINGED SIDEROBLASTS confirms sideroblastic anemia, and adequate stores with low utilization fit ACD.

BOARD TRAP

WRONG: ordering a bone-marrow biopsy when serum ferritin is already diagnostic. Marrow iron stain is reserved for ambiguous mixed cases — a low ferritin alone confirms IDA non-invasively. Don't subject a patient to marrow sampling for a question the ferritin already answered.